

Appendix D. Questionnaire

The questionnaire was administered online as a Google Form between 27 July and 2 August 2026 and is reproduced below with its wording and response options exactly as presented to respondents. Every question except the two open comment fields was required. The form carried the study’s working title from the approved proposal, “An Explainable XGBoost Framework for Predicting AI Adoption Readiness in Ghanaian Educational Institutions”; the region item (A3) offered only Greater Accra, consistent with the final scope of the study.

D.1 Informed consent (form cover page)

You are invited to participate in a research study investigating the readiness of educational institutions across Ghana to adopt AI tools in teaching, administration, and learning management.

- *Estimated time: 35–40 minutes*
- *Your institution will be assigned an anonymous identifier (e.g., INST-0042)*
- *No institution name or personal detail will appear in any publication*
- *Data stored in encrypted Google Drive with 2FA; retained 5 years post-publication*
- *Participation is entirely voluntary — you may withdraw at any time*

By submitting this form, you confirm you have read and voluntarily agree to participate.

D.2 Part A — Institutional profile

Background information about your institution. All responses are anonymised.

A1. What is your role at this institution?

- ☐ Headteacher / Principal
- ☐ Deputy Headteacher / Vice-Principal
- ☐ ICT Coordinator / IT Manager
- ☐ Dean / Director
- ☐ Registrar / Administrator
- ☐ Other
- ☐ Staff

A2. Institution Type

- ☐ Basic schools (primary + JHS)
- ☐ Senior High School (SHS)
- ☐ Polytechnic/University

A3. Region

Greater Accra

A4. Ownership Type

- ☐ Public / Government
- ☐ Private (For-profit)

A5. Approximate Student Enrolment

- ☐ Under 300
- ☐ 300 – 999
- ☐ 1,000 – 4,999
- ☐ 5,000 – 14,999
- ☐ 15,000 and above

A6. Approximate Number of Teaching Staff

- ☐ Under 20
- ☐ 20 – 49
- ☐ 50 – 99
- ☐ 100 – 299
- ☐ 300 and above

A7. Does your institution have a dedicated ICT Unit or Department?

- ☐ Yes, with dedicated full-time staff
- ☐ Yes, but staffed part-time or informally
- ☐ No

A8. Years in Current Role

- ☐ Less than 1 year
- ☐ 1 – 3 years
- ☐ 4 – 7 years
- ☐ 8 – 15 years
- ☐ More than 15 years

D.3 Part B — Adoption readiness scale

For each statement, rate how much you agree it applies to your institution RIGHT NOW. (1 = Strongly Disagree ... 5 = Strongly Agree; choose 2, 3 or 4 if you are somewhere in between.)

Table D1 lists the 35 statements. The seven marked (R) are negatively worded and were reverse-scored (6 – response) before the construct scores were computed (Section 3.5).

Table D1. Part B statements by construct, as worded in the questionnaire (R = reverse-scored).

Item	Statement	Construct
F1a	Using AI tools would improve the quality of teaching at this institution.	Perceived AI Usefulness
F1b	Using AI tools would improve student learning outcomes at this institution.	Perceived AI Usefulness
F1c	AI-powered administrative systems would increase the efficiency of our institutional operations.	Perceived AI Usefulness
F1d	Adopting AI tools would give our institution a competitive advantage in educational quality.	Perceived AI Usefulness
F1e	AI tools would make it easier for our teachers to personalise instruction for individual students.	Perceived AI Usefulness
F1f	AI tools are unlikely to make any real difference to teaching quality at our institution. (R)	Perceived AI Usefulness
F2a	Our staff would find it easy to learn how to use AI tools in their daily work.	Perceived Ease of AI Use
F2b	It would not require significant effort to integrate AI tools into our existing teaching practices.	Perceived Ease of AI Use
F2c	Adequate technical support would be available to help staff use AI tools effectively.	Perceived Ease of AI Use
F2d	Most AI tools designed for education are user-friendly enough for our context.	Perceived Ease of AI Use
F2e	Our staff would struggle to learn how to use AI tools in their daily work. (R)	Perceived Ease of AI Use
F3a	Other schools or institutions we are aware of are actively adopting AI tools.	Social and Institutional Pressure
F3b	The Ministry of Education / Ghana Education Service expects institutions like ours to adopt AI tools.	Social and Institutional Pressure
F3c	Our community (parents, local stakeholders) expects us to incorporate technology like AI into education.	Social and Institutional Pressure
F3d	We feel no pressure from peers, government, or parents to adopt AI. (R)	Social and Institutional Pressure
F4a	Our institution has reliable internet connectivity sufficient for AI tool usage.	Infrastructure Readiness

Item	Statement	Construct
F4b	We have a stable and consistent electricity supply to support technology use.	Infrastructure Readiness
F4c	Sufficient computing devices (computers, tablets, smartphones) are available for AI use.	Infrastructure Readiness
F4d	Our institution's network infrastructure can handle the data demands of AI applications.	Infrastructure Readiness
F4e	We have adequate physical space for digital technology use.	Infrastructure Readiness
F4f	Our facilities are properly configured/equipped for digital technology use.	Infrastructure Readiness
F4g	Our internet connection is not reliable enough for AI tools. (R)	Infrastructure Readiness
F5a	Most of our teaching staff have completed formal digital literacy or ICT training.	Digital Skills and Human Capital
F5b	Our institution has at least one staff member with specialist IT or AI expertise.	Digital Skills and Human Capital
F5c	Staff at this institution are willing to learn and adopt new digital technologies.	Digital Skills and Human Capital
F5d	Students at this institution have adequate digital skills to benefit from AI-powered tools.	Digital Skills and Human Capital
F5e	Most staff are reluctant to adopt new digital technologies. (R)	Digital Skills and Human Capital
F6a	Senior leadership at this institution actively champions digital innovation and technology adoption.	Leadership and Governance Support
F6b	Our institution has a formal ICT policy or digital transformation strategy in place.	Leadership and Governance Support
F6c	Leadership decisions at our institution consistently prioritise investment in educational technology.	Leadership and Governance Support
F6d	Leadership at our institution shows little interest in digital innovation. (R)	Leadership and Governance Support
F7a	Our institution has a dedicated budget line for ICT and technology adoption.	Financial and Resource Constraints
F7b	The financial resources available to our institution are sufficient to invest in AI tools.	Financial and Resource Constraints
F7c	We could identify and access external funding (grants, partnerships) to support AI adoption.	Financial and Resource Constraints
F7d	Our institution lacks the financial resources to invest in AI tools. (R)	Financial and Resource Constraints

D.4 Part C — Current AI and technology use

Assessing the current state of technology and AI deployment at your institution.

C1. Which of the following technologies does your institution currently use? (Tick all that apply)

- ☐ Learning Management System (LMS) — e.g., Moodle, Google Classroom
- ☐ School Management Information System (SMIS)
- ☐ AI-powered tools for teaching or assessment (e.g., ChatGPT, Grammarly, adaptive tutoring systems)
- ☐ AI-powered administrative tools (e.g., automated timetabling, attendance tracking)
- ☐ Video conferencing or online learning platforms
- ☐ Digital library or e-resource system
- ☐ None of the above

C2. How would you best describe your institution's current stage of AI adoption?

- ☐ Not aware — AI adoption has not yet been considered
- ☐ Aware but no action — leadership is aware of AI tools but no steps have been taken
- ☐ Exploring — we are evaluating or piloting specific AI tools
- ☐ Early adopter — we have adopted one or more AI tools in a limited capacity
- ☐ Established — AI tools are in regular use across multiple functions

C3. What is the single greatest barrier to AI adoption at your institution?

- ☐ Lack of reliable internet / power infrastructure
- ☐ Insufficient financial resources
- ☐ Low digital skills among staff
- ☐ Lack of leadership support or institutional policy
- ☐ No awareness of suitable AI tools
- ☐ Concerns about data privacy or security
- ☐ Resistance to change among staff
- ☐ Other

C4. If your institution were to receive external support to adopt AI, which type would be most valuable?

- ☐ Financial grants or subsidies for technology purchase
- ☐ Staff training and professional development
- ☐ Technical infrastructure upgrades (internet, devices, power)
- ☐ Policy guidance from the Ministry of Education / GES
- ☐ Access to vetted, low-cost AI tools designed for African educational contexts
- ☐ Peer-learning networks with other institutions that have adopted AI

C5. On a scale of 0–10, how ready do you believe your institution currently is to adopt AI tools? (Choose 0 if NOT ready at all. Choose 10 if FULLY ready. Choose any number in between.)

0 (Not at all ready) to 10 (Completely ready)

C6. Any additional comments about AI adoption readiness at your institution? (Optional)

Open text

D.5 Part D — Context, policy and outlook

A few final questions on institutional context, policy awareness, and future outlook.

D1. How long has your institution been operating?

- ☐ Less than 5 years
- ☐ 5 – 10 years
- ☐ 11 – 20 years
- ☐ 21 – 50 years
- ☐ More than 50 years

D2. When you hear “AI tools for education”, which of the following best describes your familiarity?

- ☐ I have no knowledge of what AI tools for education are
- ☐ I have heard the term but do not know specific examples
- ☐ I am aware of AI tools (e.g., ChatGPT, adaptive learning platforms) but have not used them
- ☐ I have personally tried one or more AI tools for education
- ☐ I actively use AI tools in my professional role

D3. Has your institution received external support to improve technology infrastructure in the last 3 years?

- ☐ No — we have received no external technology support
- ☐ Minimal — one-off donations (e.g., a few computers)
- ☐ Some — infrastructure improvement (e.g., internet, devices) through a government programme
- ☐ Yes — meaningful support from an NGO, private company, or international organisation
- ☐ Yes — ongoing partnership with consistent technology support

D4. How familiar are you with the Ministry of Education's Digitalisation Policy (2020–2030)?

- ☐ I am not aware of this policy
- ☐ I have heard of it but have not read it
- ☐ I am aware of it and have read parts of it
- ☐ I am familiar with it but it has had no practical impact at my institution
- ☐ I am familiar with it and it has influenced our institutional planning

D5. What role should GES / Ministry of Education play in supporting AI adoption in schools?

- ☐ Provide clear policy direction and AI adoption guidelines
- ☐ Fund infrastructure upgrades (internet, devices, electricity) directly
- ☐ Organise and fund staff training programmes on AI tools
- ☐ Vet and recommend low-cost AI tools suitable for Ghanaian institutions
- ☐ Facilitate peer-learning networks between institutions
- ☐ All of the above — a comprehensive national AI-in-education strategy is needed

D6. Does your institution have external partnerships that support AI or technology adoption?

- ☐ No partnerships of any kind
- ☐ A relationship with a university or research institution
- ☐ A partnership with a tech company or startup
- ☐ Support from an NGO or international development organisation
- ☐ Multiple active partnerships across different sectors

D7. In 5 years, where do you see AI fitting into the educational life of your institution?

- ☐ I do not see AI being relevant to our institution in 5 years
- ☐ AI may be used by a few individual staff but not institutionally
- ☐ AI tools will be piloted in one or two subject areas
- ☐ AI will be integrated into several teaching and administrative functions
- ☐ AI will be a core part of how we teach, assess, and manage the institution

D8. What aspect of AI adoption in Ghanaian education do you feel is most overlooked or underappreciated?

- ☐ The electricity and infrastructure challenge — it is more severe than policy makers realise
- ☐ Teacher resistance and the need for change management support
- ☐ The lack of AI tools designed specifically for African or Ghanaian educational contexts
- ☐ The digital skills gap among students, not just staff
- ☐ The absence of data privacy frameworks and ethical guidelines
- ☐ The widening gap between well-resourced and under-resourced institutions
- ☐ Other

Any final comments or observations you would like to share? (Optional)

Open text

D.6 Responses observed in the dataset

Table D2 shows, for each closed question used in the analysis, the options that respondents actually chose. Options in the questionnaire that no respondent chose do not appear.

Table D2. Response options chosen in the dataset for the closed questions used in the analysis (number of respondents in brackets).

Code	Question	Response options (n)
A1	Respondent role	Staff (241); Other (58); Headteacher / Principal (1)
A2	Institution type	Senior High School (SHS) (125); Basic schools (primary + JHS) (90); Polytechnic/University (85)
A5	Enrolment band	1,000 – 4,999 (123); 300 – 999 (89); 15,000 and above (87); 5,000 – 14,999 (1)
A6	Staff-count band	50 – 99 (122); 20 – 49 (90); 300 and above (88)
A7	ICT-unit status	Yes, with dedicated full-time staff (208); Yes, but staffed part-time or informally (92)
A8	Years in current role	4 – 7 years (212); 1 – 3 years (59); 8 – 15 years (21); More than 15 years (8)
C2	AI adoption stage (not used as predictor)	Exploring — we are evaluating or piloting specific AI tools (140); Early adopter — we have adopted one or more AI tools in a limited capacity (90); Aware but no action — leadership is aware of AI tools but no steps have been taken (70)
C3	Main barrier to AI adoption	Insufficient financial resources (211); Concerns about data privacy or security (87); Low digital skills among staff (1); Lack of reliable internet / power infrastructure (1)
C4	Preferred type of support	Staff training and professional development (240); Technical infrastructure upgrades (internet, devices, power) (60)
D1	Age of institution	More than 50 years (208); 21 – 50 years (89); 5 – 10 years (2); 11 – 20 years (1)
D2	Personal familiarity with AI tools	I am aware of AI tools (e.g., ChatGPT, adaptive learning platforms) but have not used them (203); I actively use AI tools in my professional role (89); I have personally tried one or more AI tools for education (5); I have heard the term but do not know specific examples (2); I have no knowledge of what AI tools for education are (1)
D3	External technology support received	Some — infrastructure improvement (e.g., internet, devices) through a government programme (273); Yes — meaningful support from an NGO, private company, or international organisation (26); Yes — ongoing partnership with consistent technology support (1)
D4	Awareness of national policy	I am aware of it and have read parts of it (130); I am familiar with it and it has influenced our institutional planning (83); I have heard of it but have not read it (78); I am familiar with it but it has had no practical impact at my institution (9)
D5	Preferred government priority	All of the above — a comprehensive national AI-in-education strategy is needed (289); Fund infrastructure upgrades (internet, devices, electricity) directly (10); Organise and fund staff training programmes on AI tools (1)
D6	Current partnerships	Support from an NGO or international development organisation (197); Multiple active partnerships across different sectors (88); A relationship with a university or research institution (13); A partnership with a tech company or startup (2)

Code	Question	Response options (n)
D7	Expected role of AI in five years	AI will be integrated into several teaching and administrative functions (127); AI will be a core part of how we teach, assess, and manage the institution (119); AI tools will be piloted in one or two subject areas (52); AI may be used by a few individual staff but not institutionally (2)
D8	Greatest concern about AI in education	The widening gap between well-resourced and under-resourced institutions (125); The absence of data privacy frameworks and ethical guidelines (90); The digital skills gap among students, not just staff (71); Teacher resistance and the need for change management support (10); The lack of AI tools designed specifically for African or Ghanaian educational contexts (4)